



ANALYSTLAB AFRICA

Experience Lab Internship

Programme Week 1 Assignment

FinTrust Digital Bank: Understand & Plan

Data Analytics Track

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FinTrust Digital Bank: Strategic Business & Data Intelligence Plan

Firm Overview

FinTrust Digital Banking is an innovative, growing financial institution operating as a retail banking and digital technology, which is built to serve customers through mobile and online banking channels. As its customer base grows, FinTrust handles massive daily volume of both customer and transaction data. To maintain its competitive and strategic advantage, the firm is transitioning from a reactive operations model to a proactive, actionable business intelligence and insightful digital solutions.

This transition will utilize an intensive advanced data analytics, machine learning and generative AI to improve transaction behavior, reduce risk, simplify operational workflows, and enhance customer support interfaces.

Strategic Objectives

This Strategic Business & Data Intelligence Plan establishes the analytical foundation required to transform raw customer and transaction data into actionable commercial insights. It serves as a master plan for the Data Analytics Track during this four(4) weeks engagement.

The following objectives will be set out to be achieved within this 4-weeks:

- **Breakdown Complex Datasets:** Map out and track FinTrust's transaction and customer datasets to establish a reliable analytical baseline.
- **Share Business Intelligence Openly:** Define high-impact KPIs to give management and stakeholders a transparent visibility both operational and understandable performance.
- **Identify Behavioral Risk:** Reveal transaction red flags, channel inefficiencies, and risk-review indicators to protect institutional assets.
- **Develop Interactive Dashboards:** Design straightforward wire frames that will serve as the structural layout for building robust executive dashboards in Power BI.

Data Analytics Track

Part A: Business Understanding

1. What business problem should FinTrust management try to solve?

FinTrust lacks a consolidated analytical layer to monitor customer behaviors, transaction patterns, and hidden operational risk vectors in real time. Because of these operational shortages, Management must have a very robust monitoring analytical system to fish out undetected high-risk behaviors that lead directly to customer attrition, increased operational costs and potential financial vulnerabilities and also transform the firm's data like the transaction and customer data into a reliable actionable business intelligence.

2. Data Analysis decisions to support management

Data Analytics Track will extract clarity from raw datasets, convert specific, detailed data into an easy-to-read format that management can instantly act upon in a very quick time frame.

Also we are going to intensively explore, clean, analyze and visualize the relevant datasets and help management derive a high level insights for a better business decisions.

3. What stakeholders could benefit from the analytical outputs?

Stakeholders are going to benefit from a comprehensive Data Profiling Report, a strategic KPI Definition Matrix, an interactive Dashboard Wireframe Structure, and a multi-week Analysis Roadmap.

Part B: Data Understanding (Both Datasets review)

1. FinTrust_Customer_Data.csv

- **Records:** 1,500 Customer records
- **Columns:** 12
- **Field names & Data types:**
Customer_ID [Text], Customer_Name [Text], Age [Numerical], Gender [Categorical], City [Categorical], Customer_Segment [Categorical], Account_Type [Categorical], Tenure_Months [Numerical], Digital_Engagement_Score [Numerical], Monthly_Income_Band [Categorical], Preferred_Channel [Categorical], Account_Status [Categorical].
- **Categorical Variables:** [Customer_ID, Customer_Name, Gender, City, Customer_Segment, Account_Type, Monthly_Income_Band, Preferred_Channel, Account_Status].
- **Numerical Variables:** [Age, Tenure_Months, Digital_Engagement_Score]
- **Relationship between the datasets:** Transactions with unknown Customer is 0
- **Missing Values:** There are zero(0) missing values in the datasets

2. FinTrust_Transaction_Data.csv

- **Records:** 12,000 Transaction records
- **Columns:** 11
- **Field names & Data types:**
Transaction_ID[Text], Customer_ID[Text], Transaction_DateTime [datetime], Transaction_Type[Categorical], Amount_NGN[Numerical], Channel[Categorical], Device_Type[Categorical], Location[Categorical], International_Transaction[Categorical], Transaction_Status[Categorical], Risk_Review_Flag[Categorical].
- **Categorical Variables:** [Transaction_ID, Customer_ID, Transaction_DateTime, Transaction_Type, Channel, Device_Type, Location, International_Transaction, Transaction_Status, Risk_Review_Flag].
- **Numerical Variables:** [Amount_NGN]
- **Relationship between the datasets:** Transactions with unknown Customer is 0
- **Missing Values:** Device_Type and Location are each missing in 96 rows

Part C: Strategic Analytical Questions

Throughout the 4-weeks, the Analytics Track is going to strategically answer a high level analytical questions which are going to help management make an inform decisions. Such analytical questions are as follows.

1. **What is the weekly distribution of transaction statuses, whether successful or failed across different channels?**

This analytical question is going to help management know how well transaction channels are performing in-terms of fast deliveries.

2. **Is there a clear statistical relationship between transaction value size and the frequency of risk review flag?**

This is actually going to answer the statistical questions like the relationships (correlations and regression), basic statistics like average values(mean), most appearing value(mode), outliers(skwness), and many more.

3. **Which customer segments generate the highest total transaction volume and financial velocity?**

This question too is going to inform management about how well customers are performing in terms of giving a large number of transactions and revenues.

4. **Do failed or reversed transactions display a heavy concentration around specific hours of the day or days of the week?**

This will also help management knows how failed or reversed transactions impact the daily operational activities.

5. **What behavioral characteristics differentiate low-risk users from those flagged for manual risk reviews?**

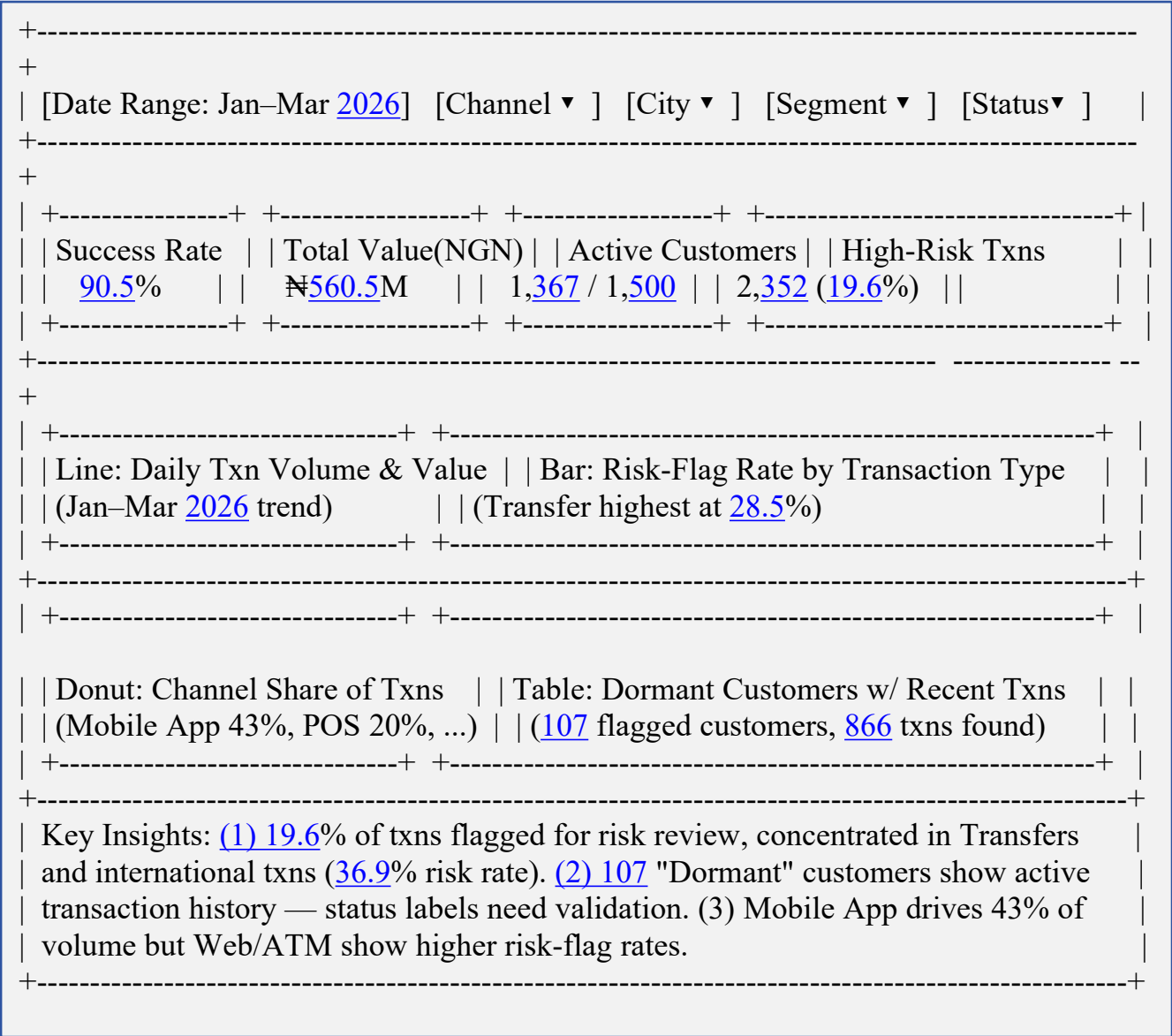
Part D: KPI Planning (6 KPIs proposed)

KPI Name	Operational Definition (How to calculate)	Strategic Decisions Support	Required Data
Transaction Success Rate (TSR)	$\frac{\text{(Successful Transactions / Total Transactions)}}{100}$	To monitor platform stability and identifies channel down-turns	Transaction_Status, Transaction_ID (<i>FinTrust_Transaction_Data.csv</i>)
Risk Exposure Index (REI)	$\frac{\text{(Flagged Risk Transactions / Total Transactions)}}{100}$	To gauge overall system risk and compliance team operational load	Risk_Review_Flag, Transaction_ID (<i>FinTrust_Transaction_Data.csv</i>)

Average Transaction Value (ATV)	Total Transaction Currency value / Total Transaction Count	To track liquidity trends and platform value-throughput	Transaction_Amount, Transaction_ID (<i>FinTrust_Transaction_Data.csv</i>)
Channel Failure Concentration	(Failures per Channel / Total Platform Failures) x 100	To highlight technical infrastructure delays in specific APIs	Channel_Type, Transaction_Status (<i>FinTrust_Transaction_Data.csv</i>)
Active Customer Transacting Base	Counts of unique Customer IDs with ≥ 1 transaction in a timeframe	To Measure active user engagement and churn patterns	Customer_ID, Transaction_Timestamp (<i>FinTrust_Transaction_Data.csv</i>)

Part E: Executive Dashboard Planning (Wireframe Dashboard)

The Production Power BI Dashboard will follow a clean, grid-based executive layout to maximize scannability:



Initial Plan for week 2 - week 4

Weeks	What is expected to achieve	How to do it	Tools to be use
Week 2	Data intensive cleaning and exploratory Analysis	The core focus of week 2 is moving from raw, synthetic data files into a verified, clean, and reliable data schema.	Excel, SQL and Python
Week 3	Relational Modeling & KPI Logig Development	The cor focus of Week 3 is structuring the internal data relationships and writing the analytical formulas that power the executive visual summaries	Excel, SQL and Python
Week 4	Dashboard Polish, Product Integration & Project Delivery	The core focus of Week 4 is refining visual design, ensuring cross-track integration alignment, and deploying the finalized Project	Power BI, Microsoft word and PowerPoint Slides

Project Success Criteria

To ensure our data analytics solution delivers true strategic solution to FinTrust, success will be evaluated across three core areas: analytics and engineering quality, business adoption, and infrastructure performance.

- **Data Reconciliation Accuracy:** The Power BI semantic layer and all analysis will 100% match with the production data base records. Calculated metrics like transaction value, counts, customer statuses and its respective dashboarding cannot diverge from raw datasets.
- **Executive Dashboard Adoption:** Success requires active usage by key business leads like Operations, Risk, and Product team). The tools used will become a primary driver in monthly performance audited and operational updates.
- **System Performance Speed:** The system user- interface will run smoothly without delay. Whiles Visual components, drill-downs, and global context filters will also process and update within a 2-second

window during active sessions.

- Operational Optimization lift: The analytics dashboard will successfully uncover hidden system inefficiencies. It will isolate at least two major processing constraint or high-failure channel routes within the first month of release.

Initial Project Risk Assessment Matrix

This assessment matrix will identifies immediate technical and delivery threats to the four-week timeline, along with practical steps to prevent them from derailing progress. This assessment matrix will contain the following:

1. Data Inconsistency and Missing Fields
2. Dashboard Interface Gap and Slow Visual Loading.
3. Scope Creep from Shifting Executive Demand
4. Cross-Track Integration